

CallEEG: Towards Cognitive Engagement-Aware Video-Mediated Communication for Copresence

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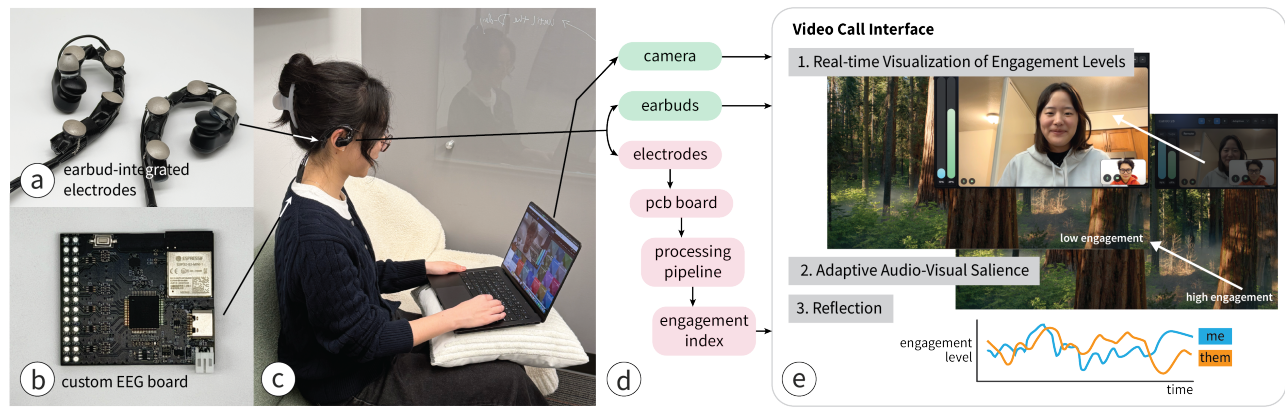


Figure 1: We propose CallEEG, which integrates an earable EEG-derived cognitive engagement index into video-mediated communication to support mutual awareness and enhance copresence. We developed (a) earbud-integrated electrodes and (b) a custom PCB for signal acquisition. During (c) a CallEEG video call, (d) EEG signals are transmitted from the PCB to a host computer, where they are processed to compute an engagement index. The engagement index is then transmitted to the remote user alongside the audio and video streams. (e) CallEEG provides three UI features: (1) real-time visualization of engagement levels, (2) adaptive audio-visual saliency, and (3) post-call reflection.

Abstract

While video-mediated communications are now a central medium for work, learning and social interaction, they provide limited social and contextual cues compared to physical interactions, weakening copresence, the sense of being together with another person. We present CallEEG, an end-to-end system that integrates ear-EEG sensing and real-time cognitive engagement inference into video-mediated communication. CallEEG contains (1) 8-channel EEG sensing earbuds with a custom PCB board, (2) a real-time data processing pipeline that derives the user's engagement level, and (3) a video communication interface with adaptive audio-visual saliency for engagement-aware communication facilitation. We validated CallEEG's engagement level processing pipeline with EEG data collected from 11 users. In addition, we conducted a 14-participant user

study to evaluate the copresence enhancement of CallEEG under various interaction levels. The results indicate that integrating cognitive engagement in video-mediated communications enhances self- and mutual awareness and ease of awareness.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; **Interactive systems and tools**;

Keywords

telepresence, cognitive engagement, EEG, video-mediated communication



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1 Introduction

Although video-mediated communication has become a central medium for work, learning, and social interaction, they provide a limited set of social and contextual cues, weakening copresence—the sense of being together with another person in a mediated environment [2]. Copresence involves not only perceiving another person, but also sensing access to their intentions, responsiveness, and internal state [4, 22]. Prior work has enriched observable cues through gaze awareness [7], simulated eye contact [24], spatialized layouts [9], embodied conferencing [8, 14], and mixed-reality remote collaboration [1]. However, users must still infer a remote partner’s availability and attentiveness using their own strategies based on audio-visual behavior.

We argue that cognitive engagement is a missing critical dimension for copresence in video-mediated communication. Although biosignal sharing has been explored for social feedback [10, 16, 21, 26], prior systems primarily capture affective or arousal-related states. Electroencephalography (EEG) is a promising modality for capturing cognitive activity [12], but is rarely integrated in social settings as conventional EEG systems are cumbersome, visually obtrusive, and difficult to incorporate into social interaction. Building on recent advances in near-ear EEG [29], we present Cal EEG, an end-to-end system that integrates earable EEG sensing, real-time engagement estimation, and an engagement-aware video communication interface to support mutual awareness and copresence.

2 System

Cal EEG is an engagement-aware video communication system that integrates near-ear EEG sensing with a real-time communication interface. The system combines earable electrodes, an EEG acquisition board, a signal processing pipeline, and a video communication interface designed to support awareness and adaptive interaction informed by EEG-derived engagement signals.

2.1 Earable Hardware and Signal Processing

We designed our earable electrodes on top of a commercial Bluetooth earbud and placed soft dry electrodes¹ in-ear and around the ear to capture EEG signals (Fig. 1(a)). 5 electrodes are placed on each side of the earbud, with one electrode on the left used as the reference and one on the right used as the bias. EEG signals are acquired from the earable electrodes using a custom bioamplifier board with ADS1299 (Fig. 1(b)). We use β/α [3, 6, 19] as our measure of cognitive engagement and calculate the metric using a signal-processing pipeline adapted from prior work [5, 13, 27, 28]. Specifically, we use 1 s epochs with a 0.25 s step size and extract band power in the alpha (7–11 Hz) and beta (11–20 Hz) range with an exponentially weighted moving average with a half-life of 5 s.

2.2 Engagement-Aware Features for VMC

Cal EEG integrates an EEG-derived engagement index into a video communication interface to support mutual awareness, adaptive interaction, and post-call reflection, inspired by prior work on biosignal-based social feedback, cognitive-load-aware interface design, and communication reflection (Fig. 1(e)).

Real-time Engagement Visualization. Cal EEG provides real-time visualizations of engagement levels of the user and their remote partner. Many works [11, 17, 18, 20] have explored how sharing biosignals can foster better understanding of each other’s state and promote a sense of connection. In Cal EEG, we leverage the EEG-derived engagement index, which may assist users in building mental models of one another during the interaction. Seeing a partner’s engagement level, while also being aware that one’s own engagement level is shared, may enhance copresence.

Adaptive Audio-Visual Salience. EEG-derived engagement signals enable the video communication interface to adapt to users’ implicit cognitive states. Prior work on context-aware adaptation of AR interfaces has shown the effectiveness of cognitive-load adaptive user interfaces [15]. Similarly, Cal EEG adjusts the salience of the conversational partner through audio spatialization, visual brightness, and UI size. For example, high engagement with low speech activity may indicate that the user is focused on an outside task, prompting the system to reduce salience and minimize distraction. Conversely, low engagement may indicate conversational availability, prompting increased audio-visual salience. These interventions represent an initial design space to be refined through future co-design with users.

Reflection. After the call, users can review their own and their partner’s engagement and speech levels to reflect on the communication dynamics of the session. Building on prior work on reflection in social and professional interactions [23, 25], we provide synchronized time-series visualizations of self and partner engagement and speech as a way to support reflection on the call experience.

3 Preliminary Evaluation

3.1 Technical Validation

We validated whether Cal EEG’s engagement index and hardware setup differentiates between tasks with different cognitive demands. 11 participants completed a high-engagement Stroop task and a low-engagement passive short-video viewing task. Across participants, the engagement index was significantly higher during the Stroop task ($M = 0.715$, $SD = 0.361$) than during short-video viewing ($M = 0.428$, $SD = 0.158$). A paired-samples t-test showed a significant difference, $t(10) = 4.09$, $p = .0022$, suggesting that earable EEG can capture task-dependent differences in cognitive engagement.

3.2 User Study

We conducted a within-subjects study with 14 participants (7 dyads) across three video-mediated activities with varying collaboration demands. Across questionnaire items measuring mutual awareness and copresence on a 10-point scale, a linear mixed-effects model showed that Cal EEG produced higher ratings than the baseline interface ($M = 6.09$ vs. $M = 5.05$; $\chi^2(1) = 29.23$, $p < .001$), indicating improved awareness-related copresence. Follow-up comparisons showed that this improvement was strongest during parallel activities ($\Delta = 1.54$, $p < .001$) and mixed activities ($\Delta = 1.17$, $p < .001$), while the difference was not reliable during shared activities. Interview responses suggested that participants used engagement cues as supplementary signals for judging partner availability and

¹<https://www.biopac.com/product/soft-pulse-kit/>

deciding when to initiate interaction. Participants also suggested additional features, such as partner-driven escalation cues.

4 Future Work and Conclusion

We presented CalIEEG, an end-to-end system that integrates ear-EEG sensing and real-time cognitive engagement estimation into video-mediated communication. We developed an earable EEG platform and an engagement-aware video call interface that supports real-time engagement awareness, adaptive salience, and post-call reflection. Findings suggest that cognitive engagement can serve as a useful signal for supporting mutual awareness and copresence, especially in video calls where users shift between individual work and active interaction. In future work, we aim to explore how users want engagement-aware features to be designed and controlled through participatory design. We also aim to understand how users appropriate EEG-derived engagement cues over time and incorporate them into everyday communication practices.

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